giant larva more than twice the weight of a normal adult. No good for propagating beetle genes, but a cornucopia for *Nosema* parasites. Giantism in beetle larvae is an extended phenotypic effect of protozoan genes.

And here is a case history to provoke even more Freudian anxiety than the Peter Pan beetles—parasitic castration! Crabs are parasitized by a creature called *Sacculina. Sacculina* is related to barnacles, though you would think, to look at it, that it was a parasitic plant. It drives an elaborate root system deep into the tissues of the unfortunate crab, and sucks nourishment from its body. It is probably no accident that among the first organs that it attacks are the crab's testicles or ovaries; it spares the organs that the crab needs to survive—as opposed to reproduce—till later. The crab is effectively castrated by the parasite. Like a fattened bullock, the castrated crab diverts energy and resources away from reproduction and into its own body—rich pickings for the parasite at the expense of the crab's reproduction. Very much the same story as I conjectured for *Nosema* in the flour beetle and for the fluke in the snail. In all three cases the changes in the host, if we accept that they are Darwinian adaptations for the benefit of the parasite, must be seen as extended phenotypic effects of parasite genes. Genes, then, reach outside their 'own' body to influence phenotypes in other bodies.

To quite a large extent the interests of parasite genes and host genes may coincide. From the selfish gene point of view we can think of *both* fluke genes and snail genes as 'parasites' in the snail body. Both gain from being surrounded by the same protective shell, though they diverge from one another in the precise thickness of shell that they 'prefer'. This divergence arises, fundamentally, from the fact that their method of leaving this snail's body and entering another one is different. For the snail genes the method of leaving is via snail sperms or eggs. For the fluke's genes it is very different. Without going into the details (they are distractingly complicated) what matters is that their genes do not leave the snail's body in the snail's sperms or eggs.

I suggest that the most important question to ask about any parasite is this. Are its genes transmitted to future generations via the same vehicles as the host's genes? If they are not, I would expect it to damage the host, in one way or another. But if they are, the parasite will do all that it can to help the host, not only to survive but to reproduce. Over evolutionary time it will cease to be a parasite, will cooperate with the host, and may